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19 October 2026

Comparison of friction draft gear models for simulation in longitudinal train dynamics

Yichang Zhou, Zhiyong Shi, Hecht M.

Abstract: The friction draft gear with hysteresis characteristics is one of the typical nonlinear elements in longitudinal train dynamics. Due to nonlinear hysteresis of friction damping, the discontinuity between loading and unloading envelopes becomes a mathematical integration issue when adopting look-up table methods to establish draft gear models. It might cause integral diverges and wrong results if being ignored. To solve this discontinuous problem and pursue a proper transitional approach, three different transitional strategies (velocity-dependent, intermediate slope, and smoothing approximation) are applied to building the draft gear model, and the one-dimensional (1D) longitudinal train model adopting the modified Newmark-ß approach, also called 'Zhai' method, is established. In order to ensure accurate results, the international benchmarking cases of longitudinal train dynamics (LTD) are studied, and dynamic responses for heavy haul trains during traction and dynamic braking are compared adopting three draft gear models with different transitional strategies in this paper. The results show that three models can successfully work with Zhai's integral method in certain step-sizes to solve longitudinal impulse problems. Compared to the other two models, the model with the velocity-dependent approach needs a minimum step (0.1 or 0.2 ms) to achieve convergent results, and its average time-costing is approximately eight times as much time as others. As this method is sensitive to the velocity difference of adjacent wagons, switching responses can change sharply, and it might cause instabilities in one simulation process. It is recommended that adopting the intermediate slope method or smoothing approximation approach to avoid discontinuity problems of modelling draft gears in LTD.

Keywords: draft gear models; transitional strategies; 'Zhai' method; benchmark cases; LTD

1. Introduction

With the increase of axle load and train length, the study of longitudinal train dynamics (LTD) has been an essential element to the railway industry, especially for freight trains, and has been addressed by worldwide researchers [1]~[4]. The development of LTD includes advanced vehicle connection models of in-train forces, extended vehicle modelling into multiple degree-of-freedoms (DOFs), complex nonlinear characteristics of in-vehicle components and other aspects. The wagon connection system, including draft gears and buffers, directly affects the in-train force, and is one of the most critical and notable parts for LTD. In fact, the draft gear and buffer system exhibits strong nonlinear characteristics, including the hysteresis [5], the 'locked unloading' behavior [6], and the varying-stiffness phenomena etc. Therefore, the modelling of this system becomes highly challenging.

Early models usually simplify the single wagons as mass points with one degree of freedom, thereby ignoring the connecting systems between car-bodies and bogies [7][8]. Some other studies packaged the draft gear into an equivalent simple mass-spring or wedge-spring, failing to represent the realistic characteristics of corresponding components as well. A refined model was proposed by Cole[6] based on dropping hammer test data, which presented the varying stiffness as deflection. The proposed model incorporated the wedge angles and static-dynamic friction and was fitted for

reproducing the 'locked unloading' behavior. An alternative is the look-up table method [9]~[11], which could reproduce the hysteresis behaviour and improve the computation efficiency, therefore is widely applied. However, the nonlinear hysteresis entails the discontinuities between loading and unloading envelopes when applying the look-up table approaches. To fix the discontinuities, smoothing approximations or transitional characteristics are generally introduced as a mathematical solution [13]. A recent review paper by Wu on the dynamics modelling of the friction draft gear [5], summarized the dynamics modelling from different institutions comprehensively, and addressed the importance of including the velocity-dependent friction, slack, limiting stiffness and transitional characteristics into draft gear models. Afterwards, productive investigations on advanced draft gear models have been published by Wu et al. [12].

Among all existing draft gear models, it is necessary to understand the applicability thoroughly before adapting them into LTD simulations. The authors compared three representative draft gear models with different transitional strategies to investigate different methods for modelling the draft gears. The simulated scenarios are referred from the international benchmarking of LTD simulators in [15][16]. This paper is organized starting from a detailed introduction on three different transitional strategies for draft gear modelling, following by a presentation of LTD model developed in MATLAB adopting the 'Zhai' method [22] for numerical integration. Section 4 includes the simulated results from four benchmark cases using the developed LTD model and comparative discussion on different methods. The conclusion and outlook are shown in Section 5.

2. Draft gear models

Center-buffer couplers and buffer-hook couplers are two types of coupling connection systems for freight wagons in the world's railway system. And Center-buffer couplers are popularly used in America, China [23][24] and Australia [5] while buffer-hook couplers are widely applied in the European railway system [25]. Early draft gear models for both types of couplers are simplified as a combination of linear stiffness and damping, ignoring the nonlinear hysteresis characteristics of the friction damping. Recent developments have incorporated different considerations into advanced draft gear models, widely using the look-up table approach.

A look-up table approach generally includes fixed tables to represent the force-deflection features of draft gears from tests, and adopts different simplifications to define the transitional strategies. As the force-deflection characteristic is dependent on the type of draft gear, the transitional characteristics affect the performance of a draft gear model significantly. To investigate the performance of different modelling methods, three draft gear models with different transitional strategies are compared in this part.

2.1 Model 1 with velocity-dependent transitional characteristics

The first transitional method in the draft gear model 1 is called a velocity-dependent transitional approach [17], and it stems from the spring and damper characteristics with a hysteresis function, as shown in Figure 1.

The draft gear is simplified as a parallel model of a linear spring and a frictional damper. The buffer force is given as the sum of the function of the relative displacement for spring functionality, and function of the relative velocity for the damper functionality. And it is a simple hysteresis force

with the upper envelope for positive relative speed and the lower envelope for negative relative speed. Since the discontinuity in the coulomb frictional force causes numerical problems for time integration, a regularization velocity v_ε is introduced to allow a steady transition between one envelope and the other. The force law is given in Equation (1).

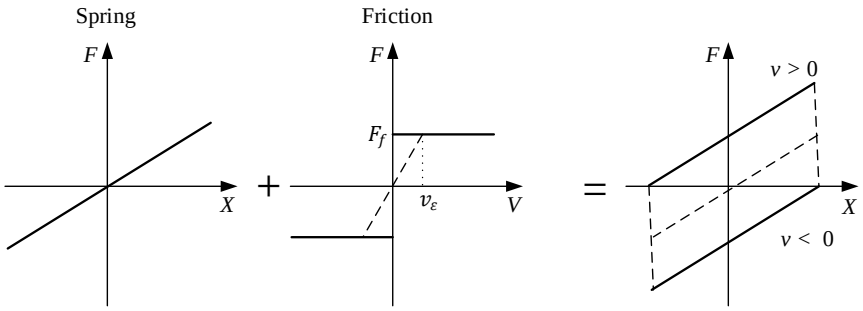


Figure 1. A simple spring and frictional damper parallel model

$$F = \begin{cases} F_{\text{spring}} \pm F_f, & |v| \geq v_\varepsilon \\ F_{\text{spring}} + \frac{v}{v_\varepsilon} \cdot F_f\,, & |v| < v_\varepsilon \end{cases} \tag{1}$$

Based on this simplified method, a detailed calculation equation for nonlinear draft gear models is proposed:

$$F(x_t, v_t) = \begin{cases} \frac{1}{2}\big(f_{\text{l}}(x_t) + f_{\text{u}}(x_t)\big) + \frac{1}{2}\big(f_{\text{l}}(x_t) - f_{\text{u}}(x_t)\big)sign(v_t \cdot x_t)\,, & |v_t| \geq v_\varepsilon \\ \frac{1}{2}\big(f_{\text{l}}(x_t) + f_{\text{u}}(x_t)\big) + \frac{1}{2}\big(f_{\text{l}}(x_t) - f_{\text{u}}(x_t)\big)sign(v_t \cdot x_t) \cdot \frac{|v_t|}{v_\varepsilon}\,, & |v_t| < v_\varepsilon \end{cases} \tag{2}$$

where x_t and v_t are relative deflections and velocities of adjacent wagons of the current time-step, respectively; f_{l} and f_{u} are the buffer loading envelope and unloading envelope, respectively; v_ε is the regularization velocity, and F is buffer force. As we can see from Equation (2) and Figure 2 (a), x_t and v_t are two critical parameters affecting buffer force while v_t also control the transitional zone.

2.2 Model 2 with intermediate slope transitional approach

The transitional method of the draft gear model 2 is original from the American railway research institutions and applied to the software package known as Train Operation Simulator (TOS) [18]. An intermediate slope is brought in to solve the interruption problem between the loading and unloading envelopes, as shown in Figure 2 (b). Thus the draft gear model can be expressed as Equation (3)-(5):

$$F(x_t) = \begin{cases} f_{\text{l}}(x_t), & x_t \geq x_{t-\Delta t} \\ f_{\text{u}}(x_t), & x_t < x_{t-\Delta t} \end{cases} \tag{3}$$

$$F(x_t) = F(x_{t-\Delta t}) + k(x_t - x_{t-\Delta t}) \tag{4}$$

$$|f_{\text{u}}(x_t)| \leq F(x_t) \leq |f_{\text{l}}(x_t)| \tag{5}$$

where $x_{t-\Delta t}$ and $F(x_{t-\Delta t})$ are relative deflections of adjacent wagons and draft gear force of the previous time-step, respectively, and k is the intermediate slope. Equation (3) determines the loading and unloading envelopes of the buffer system while the transitional characteristics between two envelopes are controlled by Equation (4) under the boundary condition (5).

2.3 Model 3 with smoothing approximation

The last transitional strategy in the draft gear model 3 is developed by the China Academy of Railway Sciences, and carried out at the software named TOSPC [19]. This smoothing approximation method was initially used to simulate the dissipating energy process during each cycle of oscillation for truck leaf springs. The draft gear model is expressed by Equation (6):

$$F(x_t) = \begin{cases} f_l(x_t) + (F(x_{t-\Delta t}) - f_l(x_t))\exp\left(\frac{-|x_t - x_{t-\Delta t}|}{\alpha}\right), & x_t \geq x_{t-\Delta t} \\ f_u(x_t) + (F(x_{t-\Delta t}) - f_u(x_t))\exp\left(\frac{-|x_t - x_{t-\Delta t}|}{\alpha}\right), & x_t < x_{t-\Delta t} \end{cases} \quad (6)$$

where α is a critical parameter to describe the rate within a hysteresis loop approaches the outer boundary of the envelop, see Figure 2 (c).

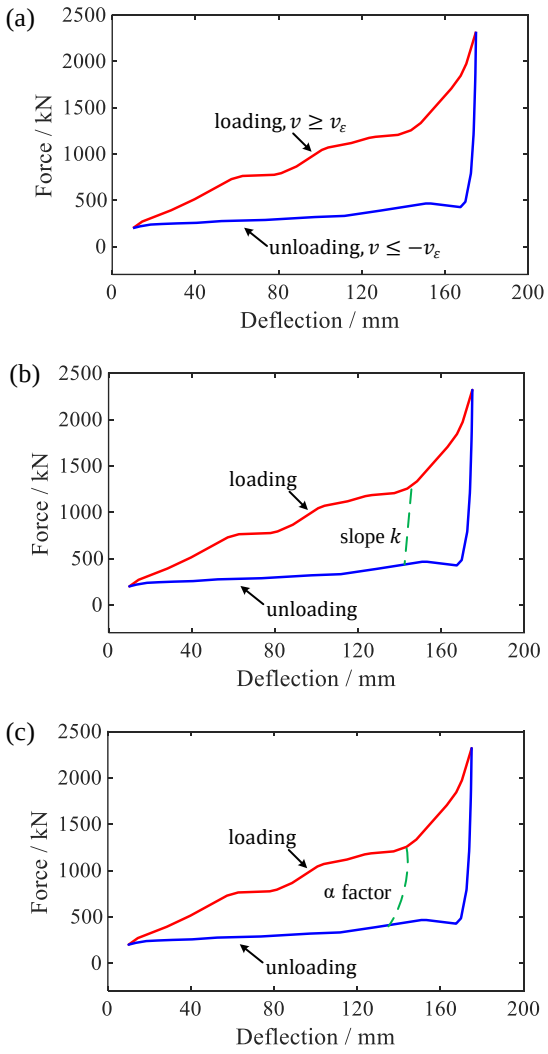


Figure 2. Different transitional approaches in three draft gear models: (a) velocity-dependent transitional characteristics in model 1; (b) intermediate slope transitional approach in model 2; (c) smoothing approximation in model 3

3. Longitudinal train dynamics model

To compare above mentioned friction draft gear models, we have applied different draft gear models into longitudinal train dynamics simulator to investigate the dynamical influence. Figure 3 illustrates one typical LTD model.

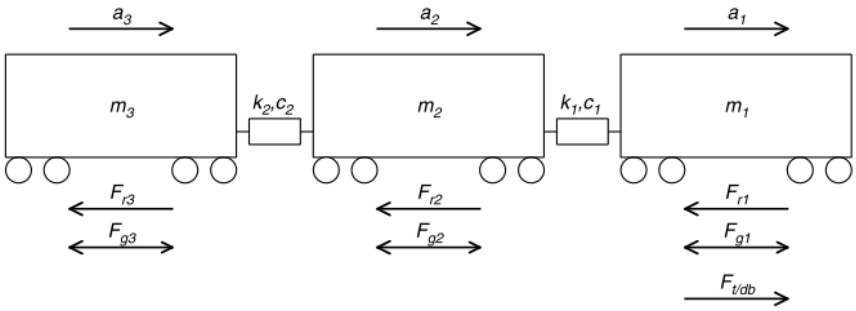


Figure 3. One typical LTD model

As shown in Figure 3, the running direction degree of freedom of each vehicle, represented by multibody dynamical components, is included. A detailed explanation regarding the model is introduced in Section 3.2.

3.1 Architecture of the LTD simulator

A general architecture of the LTD simulator is shown in Figure 4. Five parts are included, pre-processing, initialization, algorithm, net force, and post-processing. The simulator is developed in MATLAB. The first part aims to discrete input parameters, including track discretization, buffer F-D discretization, theoretical traction and dynamic braking force discretization for locomotives, and driving control sequences discretization. For example, the track curves and gradients information are interpolated by a fixed discretization step in the pre-processing stage. Without judgement or loop statements, the fixed discretization step is an effective method to save computing resources in MATLAB, and the steps of track sections, buffer-coupling system deflections, locomotive speed in the traction and DB notches, and driving control sequences are 1 m, 0.01 mm, 1 mm/s, and 1 ms, respectively [20].

The second part initializes other simulation conditions, such as the train configuration, mass matrix, locomotive and coupler positions. It should be noted that the locomotive position is relevant to traction force and dynamic braking force while the coupler position is connected with different slacks between the coupler system and draw-bar system. The third part is the core of this program including 'Zhai' integration method [22], and the displacements, velocities and accelerations of vehicles are calculated in this component. The fourth part pursues to obtain the net force, such as track resistance and running resistance for each time step, and pass the total force to the third part. The last part of this simulator is the post-processing including outputs of speed, maximum in-train force and buffer F-D envelopes.

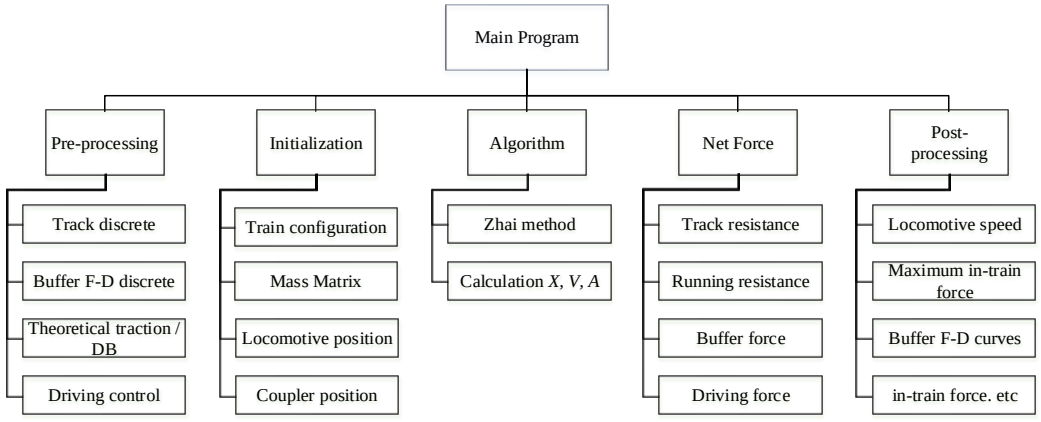


Figure 4. Architecture of the LTD model in MATLAB

3.2 Single wagon dynamics model

Two different vehicle models, the single locomotive and the single wagon are detailed in this section. The force analysis of the single wagon dynamic model is illustrated in Figure 5, and corresponding equations of motion are expressed in Equations 7 and 8.

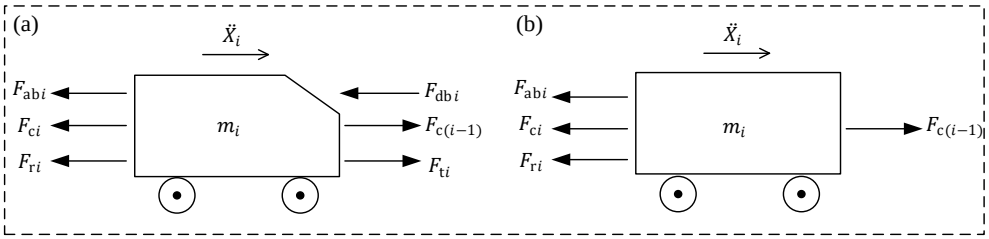


Figure 5. Single wagon dynamic model: (a) locomotives; (b) wagons

For locomotives:

$$m_i\ddot{X}_i = F_{ti} - F_{dbi} + F_{c(i-1)} - F_{ci} - F_{abi} - F_{ri} \tag{7}$$

For wagons:

$$m_i\ddot{X}_i = F_{c(i-1)} - F_{ci} - F_{abi} - F_{ri} \tag{8}$$

where m and $\ddot{X}$ are the vehicle mass and acceleration, respectively; F_c and F_{ab} are coupler force and air brake force, respectively; F_r is the sum of resistance force, including propulsion resistance, curving resistance and track slope resistance; F_{db} and F_t are the dynamic braking and traction force of locomotives, respectively; i is the number of the vehicle. All the units of variables are consistent with international standard units.

Note that the differential equations for the locomotives and wagons are different, and there is only one coupler force for the first and the last vehicles.

3.3 Numerical integration method

Newmark- ß is an approach of numerical integration to solve certain differential equations, and was developed by Nathan M. Newmark in 1959 for solving response of large-scale structural systems [14]. It is a typical implicit method with second-order accuracy, and are widely applied to dynamic systems for its high precision. However, it costs extra computing resource due to the implicit integration method. To solve this problem, Zhai [22] proposed a simple explicit method with second-order accuracy to solve large-scale nonlinear dynamic issues based on the Newmark- ß approach, and validated this new method in railway engineering. The modified method, also called 'Zhai' method, is expressed by Equation (9) and (10)

$$X_{n+1} = X_n + V_n\Delta t + \left(\frac{1}{2} + \psi\right)A_n\Delta t^2 - \psi A_{n-1}\Delta t^2 \tag{9}$$

$$V_{n+1} = V_n + (1+\varphi)A_n\Delta t - \varphi A_{n-1}\Delta t \tag{10}$$

where X_n, V_n and A_n are the approximations of the displacement, velocity and acceleration of the current time-step, respectively; Δt is the time step; ψ and φ are variables controlling the integration stability and accuracy, which are taken as 0.5 and 0.5 in this paper, respectively. As shown in Equation (9) and (10), it is a two-step method, and the displacement and velocity of the next step can be calculated according to those of the current and the previous steps. In addition, the

acceleration of each step can be given according to Equation (7) or (8), and the program is self-starting with the initial conditions X_0, V_0, F_0 and $\psi = \varphi = 0$ at the first time step.

The flow chart of the simulation for LTD is shown in Figure 6.

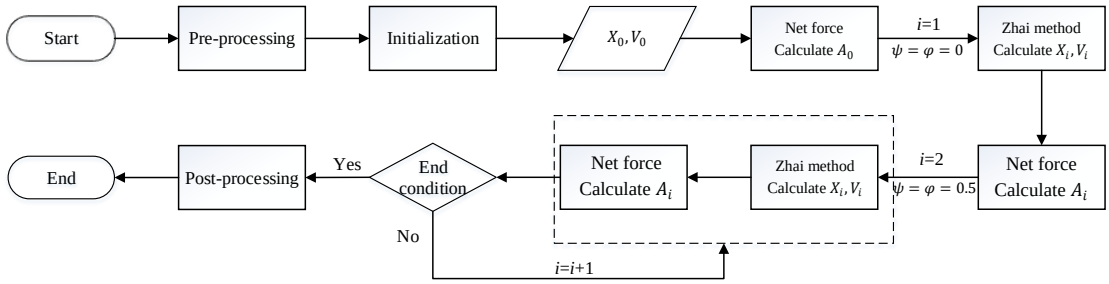


Figure 6. Flow chart of LTD simulation

4. Results and discussion

To validate the proposed LTD model, four benchmark cases, shown in Figure 7, were studied in MATLAB using a computer with an AMD FX-8350 Core at 4.00 GHz and 16 GB of RAM. For each simulation case, three draft gear models with different transitional strategies explained in Section 2 are compared.

As seen in Figure 7, four different train configurations are included: a short and a long head-end trains in Case 1 and Case 3, whereas a short and a long distributed power train in Case 2 and Case 4, respectively. It should be noticed that locomotives-1 and locomotives-2 represent two different types of locomotives while wagons-1 and wagons-2 stand for two kinds of wagons. In order to reduce the slack in the train, draw-bar connection systems were used to connect two wagons to form a wagon pair while adjacent wagon pairs were linked by coupler systems in four benchmark cases.

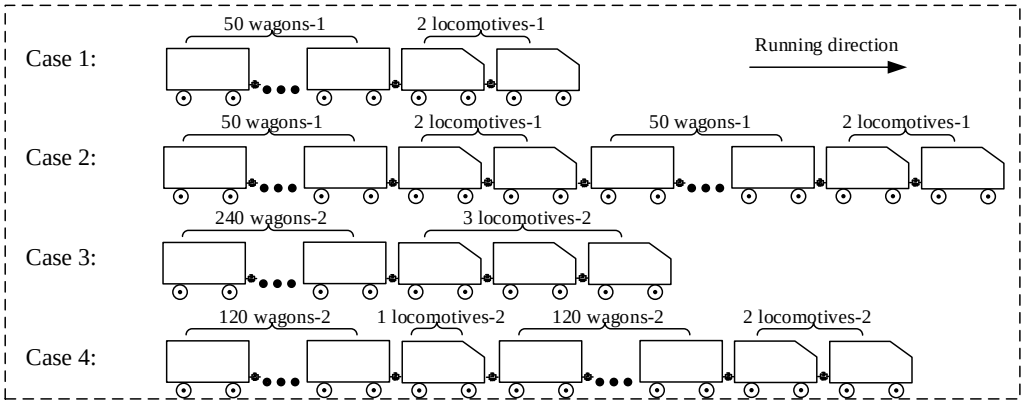


Figure 7. Train configurations of four cases

Figure 8 shows traction control strategies for each locomotive of four cases, in which traction force is represented by positive values while dynamic braking of locomotives is represented by negative values. Besides, the parameters for trains, wagon connection systems, and more input information of LTD are referred to the article [16].

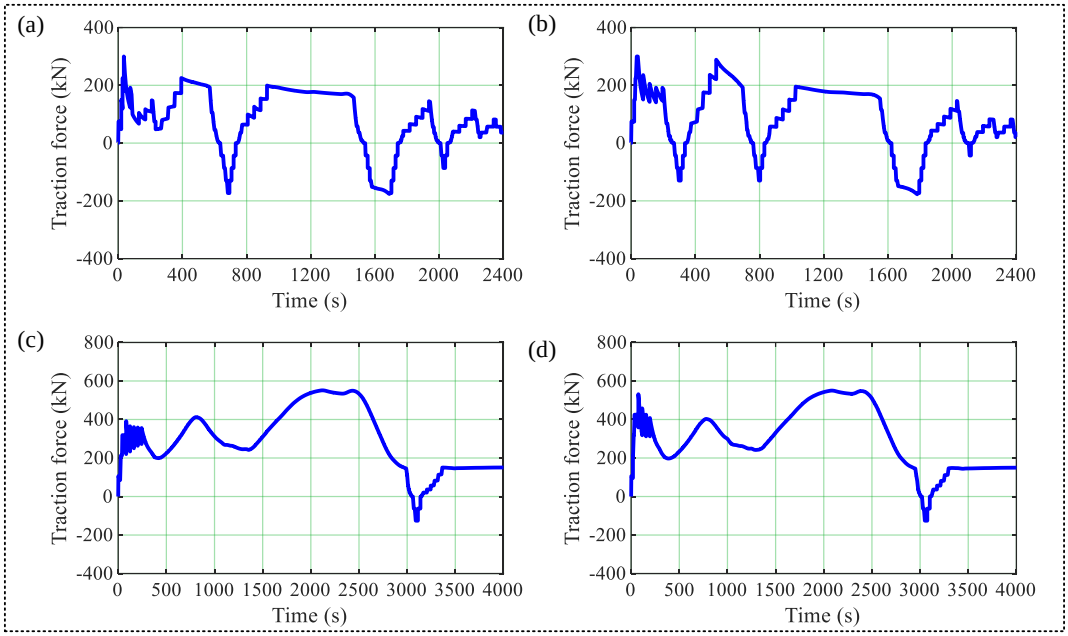


Figure 8. Locomotive traction controls under four cases: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4.

4.1 Results of CASE 1

Figure 9 shows the results of the speed of the leading locomotive, maximum longitudinal force, 10th in-train force, and buffer F-D characteristics in Case 1.

It can be seen from Figure 9 (a) that the speed of the leading locomotive synchronously changes with the location when adopting three draft gear models with different transitional characteristics. Furthermore, the speed limits in the magenta dashed line are provided to aid checking the simulation results while 10% of over-speed is allowed. For in-train forces, it is stipulated that the tensile force is represented by negative values while the compressive force is represented by positive values in this paper. As presented in Figure 9 (b), the maximum compressive and tensile in-train forces both arose between the last locomotive and the first wagon, and values of the largest forces are rather close. The variation of maximum compressive in-train forces, as shown in Table 1, is only 3 kN (from 343 to 346 kN) whereas that of maximum tensile in-train forces is 12 kN (from 590 to 602 kN). In addition, adopting model 2 had the largest average of maximum tensile and compressive forces while adopting model 1 had the lowest average forces.

Before further analysis is executed, two terms would be clarified firstly. The 10th in-train force means that the force occurred between the 10th and 11th vehicles from the train head while the 10th buffer means that the buffer is located between the 10th and 11th vehicles. In terms of the 10th in-train forces in the time domain, the response results of the three models were close, as shown in Figure 9 (c). In order to clearly present the results, the in-train forces are shifted upwards by 0, 300, 600 kN, respectively. The highest tensile forces all happened around at t = 37 s. In comparison, the maximum compressive forces arose approximately at t = 680 or 1600 s. It is evident that the whole trend of force curves are associated with traction curves.

Figure 9 (d) shows that the maximum negative and positive deflections of the model 1 are the highest above all, and the values are -71 and 49 mm, respectively. Besides, the 10th buffer F-D characteristics of the model 2 and 3 have near responses. Even though the maximum deflection of

model 1 is larger than others, the buffer force sometimes is even smaller due to the velocity-dependent switching strategy. And most buffer forces are obtained in the transition zone.

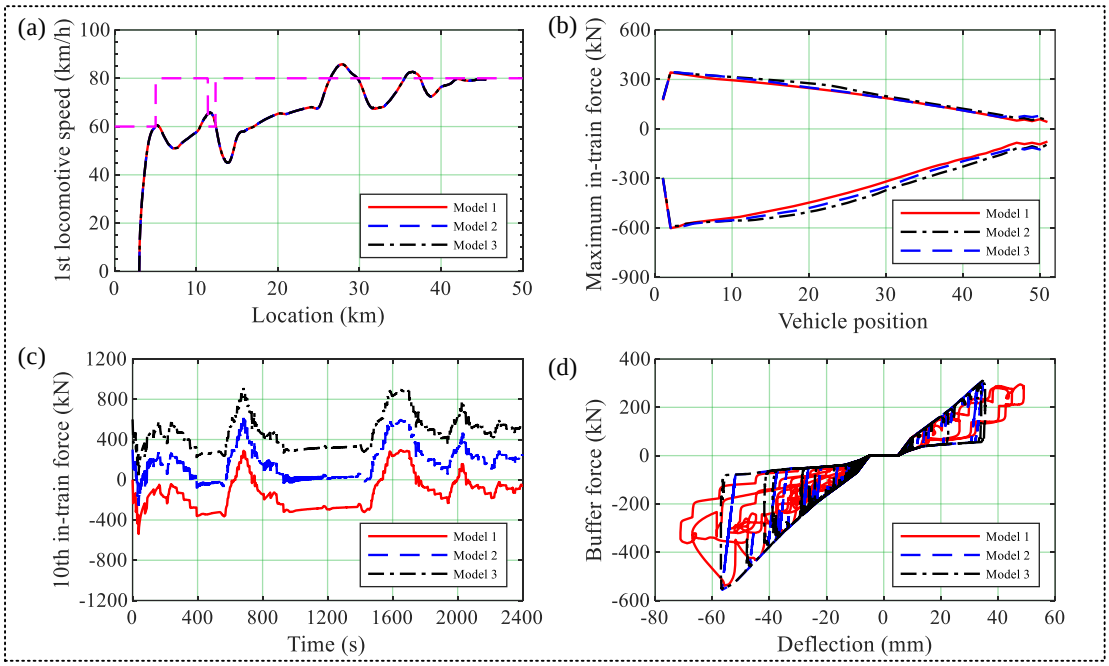


Figure 9. Simulation results of Case 1: (a) 1st locomotive speed, (b) Maximum in-train force, (c) 10th in-train force, and (d) 10th buffer force

4.2 Results of CASE 2

Figure 10 describes the results of the speed of the leading locomotive, maximum longitudinal force, 10th in-train force, and buffer F-D characteristics in Case 2.

Similar to Case 1, the speeds of the leading locomotive adopting three draft gear models for LTD simulation are almost the same, as shown in Figure 10 (a). Figure 10 (b) illustrates that three models had similar maximum force distribution along the train. The highest compressive and tensile in-train forces always arise in the process from the locomotive transferring to the wagon. Since the configuration of this distributed power train is a 'locomotive + wagon + locomotive + wagon' pattern, there are two peaks in the compressive or tensile in-train force curve. One is located between the last locomotive and the first wagon in front; the other exists between the last locomotive and the first wagon in middle. The second peak is generally higher due to the larger longitudinal impulse. As presented in Table 1, the maximum tensile in-train forces of three models are 762, 763, and 777 kN respectively, while the maximum compressive are all 418 kN. Besides, the averages of the maximum in-train forces have tiny differences applying three models.

As seen in Figure 10 (c), the dynamic responses of the 61st in-train forces approximately change synchronously in the time domain applying three draft gear models. The maximum tensile forces all happen around at t = 44 s while the largest compressive forces arise approximately at t = 1735 s. Even though the 61st in-train forces in the time domain are almost the same, the deflections of draft gears shown in Figure 10 (d) are entirely different due to the different transitional strategies adopted in three draft gear models, especially for the model 1. As described in Table 1, the maximum tensile deflections of the model 1, model 2 and model 3 are 93, 81 and 82 mm, respectively. Furthermore, the highest compressive deflections of the model 1 was 56 mm while those are only 34 and 35 mm for the model 2 and 3, respectively. In conclusion, the buffer force mainly distributed in the transition

zone of buffer F-D characteristics adopting model 1 compared other models.

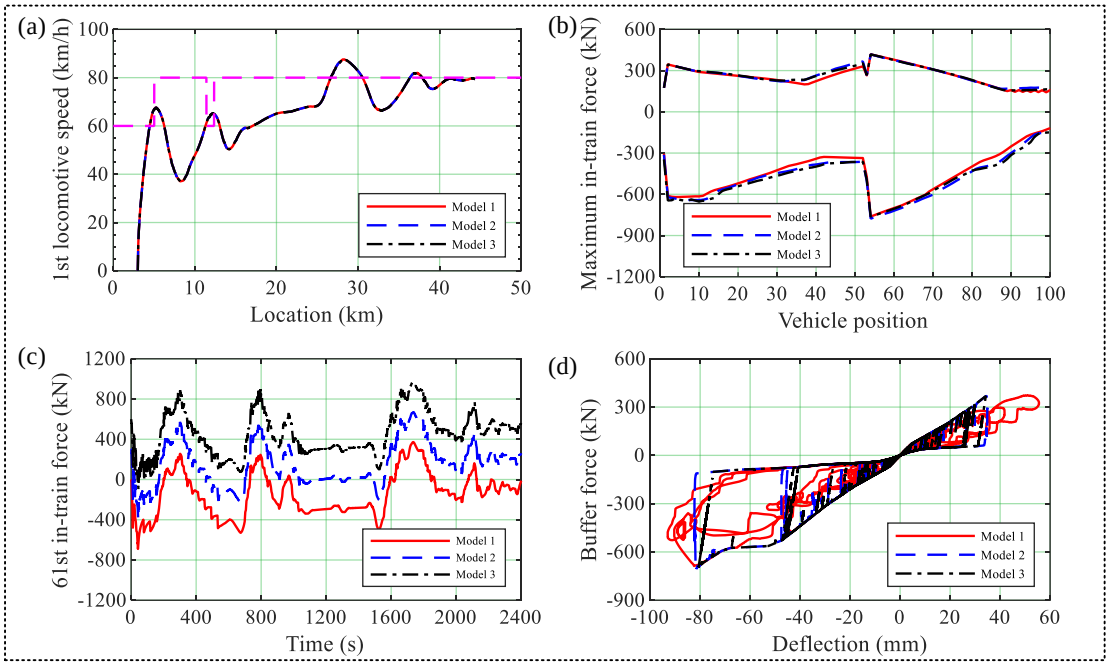


Figure 10. Simulation results of Case 2: (a) 1st locomotive speed, (b) Maximum in-train force, (c) 61st in-train force, and (d) 61st buffer force

4.3 Results of other two cases

The detailed dynamic responses of Case 3 and Case 4 are attached in Appendix 1 and 2, respectively. Similar to Case 1 and 2, the first locomotive speed adopting three models are almost the same and variations of the maximum in-train force between three models are tiny in both cases. The responses of the 147th in-train force and buffer force are calculated applying three models in Case 3 while those of the 146th in-train force and buffer force are shown in Case 4. Three models in both cases all had similar in-train force responses of the selected position in the time domain. However, the buffer F-D characteristics of the chosen place have some differences due to the different transitional strategies, especially for the model 1.

4.4 Discussion

The results from the table 1 summarize the simulation results for four cases adopting three different draft gear modelling methods. The calculated in-train force and the deflection of selected coupler position using three different draft gear models can reach well agreements with referenced results.

The results of Case 3 are similar to Case 1, whereas those of Case 4 are close to Case 2 due to similar train configurations. Compared with Case 1, the maximum compressive force occurred at nearly the three-fifth of the train length in Case 3 during train operations. The largest tensile and compressive forces in Case 2 and 4 all arose near to the locomotives in the middle of the train. Among all the cases, the average of maximum tensile and compressive force adopting model 1 is the lowest whereas the maximum buffer deflections of chosen positions are larger than others.

Table 1. Main simulation results for all four cases

Cases	Draft gear models	Max.in-train forces (kN) /coupler position	Mean of Max. in-train forces (kN)	Max. deflection of selected coupler position (mm)

		Tensile	Comp.	Tensile	Comp.	Pos.	Tensile	Comp.
Case 1	Model 1	-602/2	343/2	-348	198	10	-71	49
	Model 2	-590/2	346/2	-386	214	10	-57	35
	Model 3	-601/2	345/2	-370	205	10	-57	36
Case 2	Model 1	-762/54	418/54	-444	259	61	-93	56
	Model 2	-763/54	418/54	-476	266	61	-81	34
	Model 3	-777/54	418/54	-472	268	61	-82	35
Case 3	Model 1	1644/4	977/147	-1163	599	147	-166	133
	Model 2	1642/3	980/146	-1172	640	147	-144	120
	Model 3	1643/3	985/145	-1176	633	147	-145	121
Case 4	Model 1	-1310/123	848/122	-931	482	146	-161	105
	Model 2	-1319/123	848/122	-984	514	146	-140	79
	Model 3	-1318/123	843/122	-977	508	146	-141	80

The computing speed of all the discussed cases are listed in Table 2. The results show that the model 1with the velocity-dependent transitional characteristics requires a smaller step size than the other two approaches in each case. In other words, it demands lots of computing resource to finish the simulation adopting this method. Furthermore, model 2 and model 3 can realize online simulations during train operations.

Table 2. Computing speed of four cases

Transition methods	Cases	Step-size (ms)	Wall time (s)	Train operation time(s)	Speed 1 (ms/s)
Model 1	Case 1	0.2	809	2400	337
	Case 2	0.1	3692	2400	1538
	Case 3	0.1	10440	5000	2088
	Case 4	0.1	10458	5000	2092
	Average	–	–	–	1514
Model 2	Case 1	1	166	2400	69
	Case 2	1	365	2400	152
	Case 3	1	1292	5000	258
	Case 4	1	1263	5000	253
	Average	–	–	–	183
Model 3	Case 1	1	171	2400	71
	Case 2	1	372	2400	155
	Case 3	1	1316	5000	263
	Case 4	1	1331	5000	266
	Average	–	–	–	189

5. Conclusions

In this paper, three representative draft gear models with different transitional strategies to investigate different methods for modelling the draft gears were compared. The LTD simulator including the draft gear models were presented in detail. Based on the 'Zhai' integral method, developed in MATLAB, the simulator enables the analysis of longitudinal dynamics problems. The following conclusions can be obtained:

(1) The three transitional methods (velocity-dependent, intermediate slope, and smoothing approximation) can successfully cooperate with 'Zhai' integral method, a two-step explicit method, to solve the longitudinal impulse problems in LTD, and the results are in good agreement with the benchmark results. The dynamic responses (including results and calculating time of four cases) of intermediate slope and smoothing approximation have tiny divergences.

(2) The velocity-dependent method requires a minimum integration step (0.1 ms or 0.2 ms) to get stable responses while the step sizes of other two approaches are only 1 ms. Adopting the intermediate slope or smoothing approximation method can save more calculating resources and realize online simulations.

(3) The velocity-dependent method is sensitive with the velocity difference of adjacent wagons while other two approaches are controlled by the relative displacement of adjacent wagons. If the velocity sharply changes, it might cause instabilities in the simulation process. Hence, the intermediate slope method and smoothing approximation approach are recommended to deal with the discontinuity problems of modeling frictional draft gears in LTD.

(4) The maximum buffer deflections of the velocity-dependent method are obviously higher than others, and the highest in-train forces are obtained in the transitional zone due to its smoothing strategy. This explains the reason why its highest in-train forces are almost equal to others although the buffer deflections are obviously larger.

Apart from approaches introduced in this paper, there still remain other methods to establish the draft gear model and further comparisons are supposed to execute in the future.

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Appendices

Appendix 1

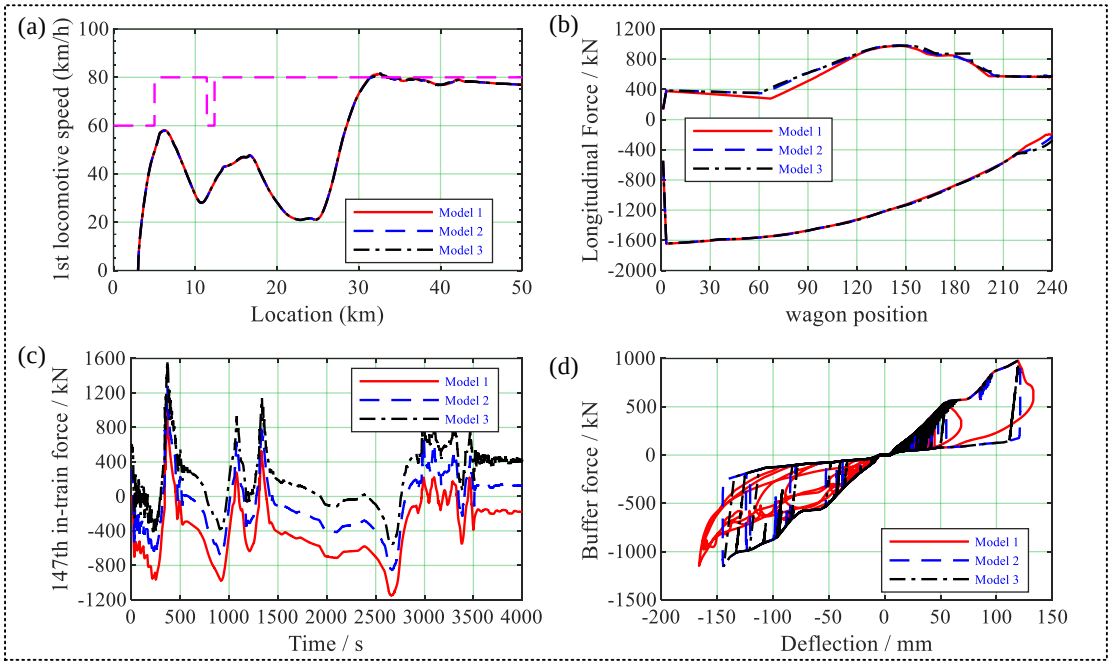


Figure 11. Simulation results of Case 2: (a) 1st locomotive speed, (b) Maximum in-train force, (c) 147th in-train force, and (d) 147th buffer force

Appendix 2

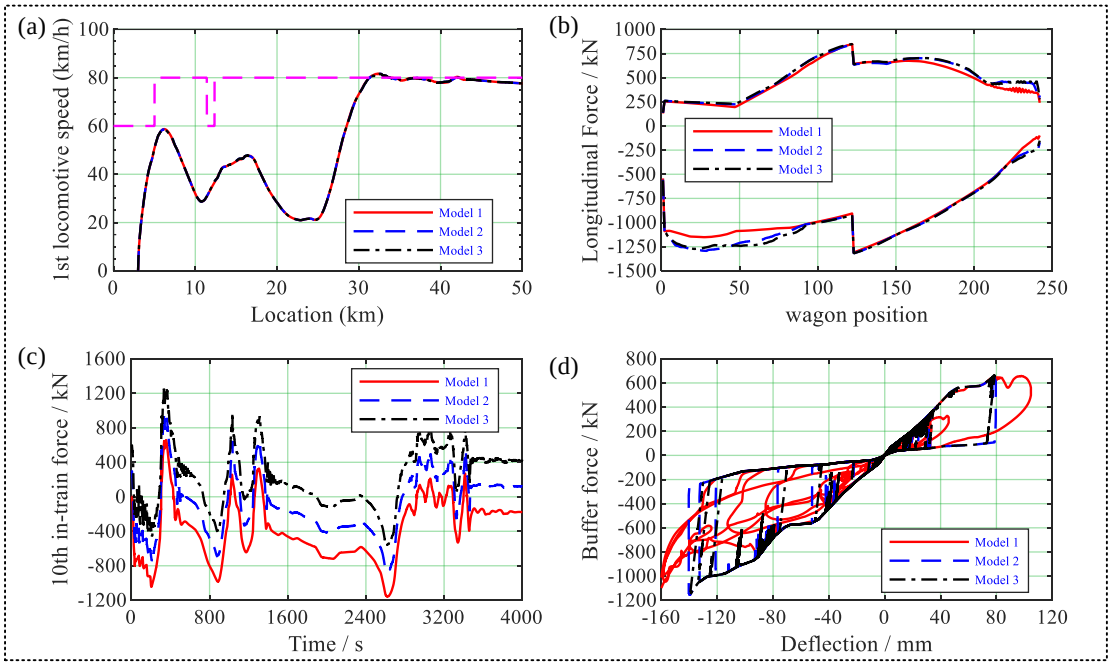


Figure 12. Simulation results of Case 2: (a) 1st locomotive speed, (b) Maximum in-train force, (c) 146th

in-train force, and (d) 146th buffer force